Matrix Models for Membranes

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Abstract

In this report we study gauged matrix quantum mechanics on a Euclidean time lattice using Hamiltonian Monte Carlo methods. We begin from the matrix harmonic oscillator and gauge its $U(N) \times SO(d)$ symmetry. We reduce the gauge data to the eigenphases of the group holonomies and incorporate them into the HMC simulation including their Faddeev-Popov actions. Replacing the Gaussian potential with a commutator squared term produces the bosonic BFSS Hamiltonian. We once again gauge its $U(N) \times SO(9)$ symmetry. We consider the projection of the bosonic BFSS system onto the orientifold, and gauge its resulting $SO(N) \times SO(9)$ symmetry. For each system we estimate critical temperatures given by the Polyakov loop susceptibility. We verify these by examining the behaviour of energy, extent of space and the Polyakov loop itself at these temperatures. Our estimates agree with known results where available, where in all cases we have specified $d = 9, N = 10$. For the Gaussian $U(N)$ model we find $T_c \approx 0.44$ in good agreement with the large-N Hagedorn value $\frac{1}{\ln(9)} \approx 0.46$. For the $U(N)$ gauged bosonic BFSS we find $T_c \approx 0.87$ which is also in agreement with the expected $T_c \approx 0.9$ for high N. Gauging the $U(N) \times SO(d)$ symmetry of these models produces large statistical errors and limits the analysis of critical temperature to a window. For the bosonic BFSS on the orientifold with the $SO(N)$ gauge we find a transition at $T_c^{\text{colour}} \approx 0.65$. Once the $SO(d)$ gauge is introduced the statistical error again becomes larger, but the critical temperatures for the $SO(d)$ and $SO(N)$ Polyakov loops coincide within error at $T_c^{\text{colour}} \approx T_c^{SO(d)} \approx 1.25$.

Contents

1 Introduction **3**

2 The Quantum Harmonic Oscillator **4**
2.1 Matrix Models 5
2.2 The $U(N)$ Gauged Matrix Model 6
2.3 The $U(N) \times SO(d)$ Gauged Matrix Model 8

3 The Bosonic BFSS Model **10**
3.1 The Orientifold 10

4 Observables **11**

5 Results **11**
5.1 The Quantum Harmonic Oscillator 12
5.2 The Bosonic BFSS 13
5.3 The Bosonic BFSS on the Orientifold 15

6 Conclusion **16**

Appendix A: Hybrid Monte Carlo Methods **18**

1 Introduction

Our current understanding of elementary particle physics is based on the Standard Model. The standard model manages to include 3 of the 4 fundamental forces: the electromagnetic, strong nuclear, and weak nuclear forces. This of course excludes gravity, which has motivated decades of research searching for a theory that can describe gravity consistently with quantum mechanics. Supersymmetry was found to be a useful tool to push our understanding further than the standard model. This led to 11 dimensional super gravity and then simultaneously to the study of 10 dimensional super strings. This left us with 5 consistent superstring theories including Type I, Type IIA, Type IIB, and Heterotic $SO(32)$ and $E_8 \times E_8$ [3]. These string theories are referred to as perturbative as interactions are controlled by the string coupling g_s in a power series. This means in cases where g_s is not small the perturbative expansion is no longer reliable, motivating a non perturbative theory.

In 1995 Witten argued that the strong coupling limit of the 10 dimensional Type IIA superstring is described by an 11 dimensional theory whose low energy limit is 11 dimensional supergravity [20], and then later it was proposed that the 5 superstring theories and 11 dimensional supergravity are in fact limiting cases of one underlying non perturbative theory. This theory is called 11 dimensional M-theory, the naming of which has been left ambiguous but is often proposed to stand for 'membrane'. A membrane is a 2 dimensional vibrating hyper-surface, similar to the top of a drum. In general, we refer to any p+1 dimensional vibrating hyper-surface in spacetime as a p-brane, and Dirichlet p-branes as Dp-branes [15, 14]. Where by Dirichlet brane we mean a brane where the endpoints of open strings are anchored, see fig.1.1a.

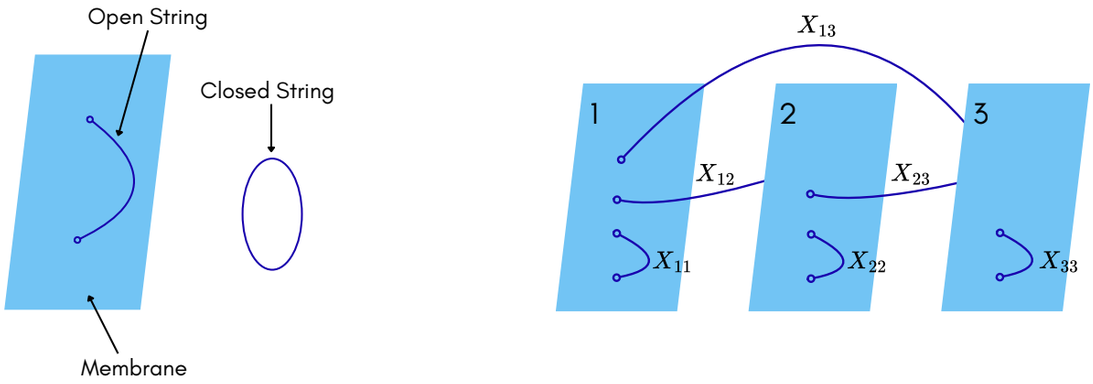
Matrix theory was first derived as a finite dimensional regularisation of 11 dimensional supermembrane theory in light-front gauge [18, 16]. It was then later conjectured by Banks, Fischler, Shenker, and Susskind (BFSS) that the $N \to \infty$ limit of supersymmetric matrix quantum mechanics describing N D0-branes is equivalent to uncompactified 11-dimensional M-theory [1]. The physics of N coincident Dp-branes at low energy can be described in static gauge by the dimensional reduction of $\mathcal{N} = 1$ Super Yang Mills (SYM) in 10 dimensions to $p + 1$ dimensions. Describing N D0-branes then is most directly done by reducing the 10 dimensional SYM down to 0+1 dimensions [2]. The resulting action of the dimensionally reduced theory is [4],

$$S_M = \frac{1}{g^2} \int dt \, \mathrm{Tr} \left\{ \frac{1}{2} \left(\mathcal{D}_0 X^i \right)^2 + \frac{1}{4} \left[X^i, X^j \right]^2 - \frac{i}{2} \Psi^T C_{10} \Gamma^0 \mathcal{D}_0 \Psi + \frac{1}{2} \Psi^T C_{10} \Gamma^i \left[X^i, \Psi \right] \right\} . \tag{1.1}$$

For our report we focus on the bosonic part of this theory, whose action is given by,

$$S_b = \frac{1}{g^2} \int dt \, \mathrm{Tr} \left\{ \frac{1}{2} \left(\mathcal{D}_0 X^i \right)^2 + \frac{1}{4} \left[X^i, X^j \right]^2 \right\} . \tag{1.2}$$

The fields X^i $(i = 1, \ldots, 9)$ are $N \times N$ Hermitian matrices that transform as a vector under the $SO(9)$ rotations of the transverse space and as scalars with respect to the one-dimensional worldvolume. The matrix structure has a simple origin. An open string can end on any of the N branes, so each endpoint carries a label $i = 1, \ldots, N$, and a string stretched from brane i to brane j gives rise to a field A^{μ}_{ij} (fig. 1.1b). The fields therefore assemble into $N \times N$ matrices whose transverse components are the scalars X^a giving the positions of the branes. A stretched string has energy proportional to its length, so the (i, j) field has a mass proportional to the separation of branes i and j. As the branes approach one another these strings become massless and, as first argued by Witten [19], the off-diagonal fields join the diagonal ones to enlarge the gauge symmetry from $U(1)^N$ to $U(N)$. For this reason we will later investigate gauging the $U(N)$ symmetry and then later look at the $U(N) \times SO(d)$ case.



(a) (b)

Figure 1.1: In figure (a) a Dp-brane is shown with an open and closed string. The open string connects its ends to the brane. In figure (b) three coincident Dp-branes are shown. The string ends are labeled, and the matrix elements of X correspond to which string end is on which brane. In the low energy limit these branes are pushed closer together.

In this report we first investigate the matrix quantum harmonic oscillator (QHO) and gauge its $U(N) \times SO(d)$ symmetry. We then show how the bosonic BFSS system can be built from the matrix QHO with the Gaussian potential term swapped for a commutator squared. We gauge the $U(N) \times SO(d)$ symmetry for this model, and then project the BFSS action onto an orientifold where we instead gauge an $SO(N) \times SO(d)$ symmetry. These actions are all discretized and allow us to use Hamiltonian Monte Carlo (HMC) methods to compute thermodynamic quantities, the details of which can be found in appendix A. We present some approximate critical temperatures for the aforementioned models where they are suspected to undergo a phase transition and compare these where possible to known results.

2 The Quantum Harmonic Oscillator

The Quantum Harmonic Oscillator (QHO) is a very well known example of a quantum system. Its Hamiltonian is given in terms of the creation and annihilation operators, a and $a^\dagger$, as,

$$H = \omega(a^\dagger a + \frac{1}{2}). \tag{2.1}$$

Then its partition function is calculated via the usual formula from statistical mechanics and has a closed solution,

$$Z = \ \mathrm{Tr}(e^{-\beta H}) = \sum_{n=0}^{\infty} \langle n|e^{-\beta H}|n\rangle = \frac{e^{-\beta\omega/2}}{1-e^{-\beta\omega}} \tag{2.2}$$

In this section we will upgrade this description firstly to the path integral formalism of quantum mechanics. In doing this, we will show how the partition function will give us a probability distribution of $P(x) \propto e^{-S_E}$, where S_E represents an action. Firstly, we begin with the definition of the Feynman pathintegral. The Feynman path integral quantifies the amplitude for a particle to evolve from a position q_i to a position q_f at a time t [17, 13],

$$\langle q_f|e^{-iHt}|q_i\rangle = \int_{q_i(0)=q_i}^{q_f(t)=q_f} \mathcal{D}q e^{iS}. \tag{2.3}$$

Where in the above relationship S denotes the classical action given most generally by,

$$S = \int_0^t dt' \left[\frac{m}{2}\left(\frac{dq}{dt'}\right)^2 - V(q)\right]. \tag{2.4}$$

Our goal is to manipulate eq.2.3 such that we can return the definition of the partition function in eq.2.2. Firstly, to turn eq.2.3 into a statistical partition function we must deal with the factor of i. This is done through a Wick rotation, which just means working with the Euclidean time variable $\tau = it$. This rotation produces the Euclidean Action S_E,

$$iS = -S_E = -\int_0^\tau d\tau' \left[\frac{m}{2}\Big(\frac{dq}{d\tau'}\Big)^2 + V(q)\right]. \tag{2.5}$$

Now, we require that the particle propagates for β Euclidean time,

$$\langle q_f|e^{-H\beta}|q_i\rangle = \int_{q_i(0)=q_i}^{q_f(\beta)=q_f} \mathcal{D}q\ e^{-S_E}. \tag{2.6}$$

Note that the trace of a given quantity x has the following form

$$\mathrm{Tr}(x) = \int dq_i \, \langle q_i | x | q_i \rangle, \tag{2.7}$$

so if we would like to return a trace from the path integral in eq.2.6 then we must require that $q_f = q_i$ and integrate over all q_i. We can achieve this by enforcing periodic boundary conditions on Euclidean time such that $\tau = \tau + \beta$. With these two changes it's clear we reproduce the desired result,

$$Z = \mathrm{Tr}(e^{-\beta H}) = \int dq_i \langle q_i | e^{-H\beta} | q_i \rangle = \int dq_i \int_{q_i(0)=q_i}^{q_f(\beta)=q_f} \mathcal{D}q \; e^{-S_E}. \tag{2.8}$$

We've shown how the statistical Z can be built from the path integral. We then have two different formulations for the partition function in the contexts of statistical mechanics and quantum field theory. Now we will divert our attention back to the specific case of the QHO. In this case we define a position coordinate of $q = x$, and we have a Euclidean action of,

$$S_E = \int_0^\beta d\tau \frac{1}{2}[(\partial_\tau x)^2 + (\omega x)^2], \tag{2.9}$$

where ω is just a constant. We seek to discretize this action to make it solvable on a computer, so it is necessary to describe this action on the Euclidean time lattice. We introduce a periodic lattice of Λ points with spacing $a = \frac{\beta}{\Lambda}$ between them. With this construction, $\tau \to \tau_n = na$, the action becomes a sum [11],

$$S_E = \sum_{n=1}^{\Lambda} \frac{a}{2}[(\partial_\tau x_n)^2 + (\omega x_n)^2]. \tag{2.10}$$

Where we have not yet discretised the derivative. Then, we can rewrite $(\partial_\tau x_n)$ as,

$$\partial_\tau X \to \frac{X_{n+1} - X_n}{a} = \frac{e^{a\partial_\tau} - 1}{a}. \tag{2.11}$$

The construction of moving to a periodic time lattice is often best understood visually, as shown in fig.2.1. Euclidean time has been parametrised as a circle of period β with defined on it Λ discrete points. In the following discussions we will continue to discretise actions on the lattice in the way we have shown here.

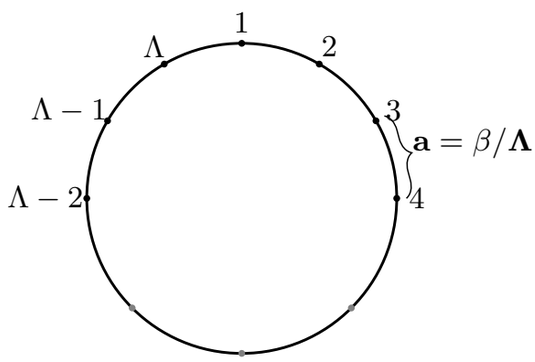


Figure 2.1: Discretisation of the Euclidean time circle into Λ lattice sites.

2.1 Matrix Models

We must extend our discussion to include matrices in order to approach our goal of using the BFSS conjecture. In ordinary quantum mechanics we describe a system by $x(t), p(t)$ and their respective commutation relations. Now, we consider $x(t), p(t) \to X(t), P(t)$ where $X(t), P(t)$ are both $N \times N$ matrices. For the purposes of our discussion these matrices can be understood as the configurations of attaching open strings to $ND0$ branes. We consider d $N \times N$ matrices at each lattice site. We denote each of these matrices with X_n^i where $n = 1, 2, \ldots, \Lambda$, and $i = 1, 2, \ldots, d$. For example, in the case of the bosonic BFSS model $d = 9$, and in the supersymmetric BFSS model there are 16 more fermionic degrees of freedom. In this report we have focused on the bosonic BFSS model, so in general we consider $d = 9$. Despite the variables becoming matrices the action of course must remain a scalar. A natural way to construct such a scalar from matrix valued quantities is by taking the trace. Then the Euclidean action is,

$$S = \frac{1}{2} \int_0^\beta d\tau \mathrm{Tr} \left[\sum_{i=1}^{D} (\partial_\tau X^i)^2 + (\omega X^i)^2 \right]. \tag{2.12}$$

Now we discretise this action as before, at which point it is straightforward to implement numerically.

$$S = \frac{a}{2} \sum_{n=1}^{\Lambda} \sum_{i=1}^{d} \mathrm{Tr} \left[(\partial_\tau X^i)^2 + (\omega X^i)^2 \right] = \frac{a}{2} \sum_{n=1}^{\Lambda} \sum_{i=1}^{d} \mathrm{Tr} \left[\left(\frac{X_{n+1}^i - X_n^i}{a} \right)^2 + (\omega X_n^i)^2 \right] \tag{2.13}$$

The introduction of the trace and matrix valued X_n^i has given us some global symmetries. In the following sections we will localise these symmetries by gauging them appropriately and introducing a covariant derivative.

2.2 The $U(N)$ Gauged Matrix Model

In this section we consider the case where $d = 1$ for convenience and then generalise our results at the end. We try to go over the main methods we used for gauging matrix systems in detail. The following sections in which gauging is mentioned will skip many of these steps and may refer back to what we discuss here.

The trace is cyclic and as such we have the freedom to make the transformation $X \to UXU^{-1}$, since $\mathrm{Tr}(UXU^{-1}) = \mathrm{Tr}(XU^{-1}U) = \mathrm{Tr}(X)$, for some U an $N \times N$ invertible matrix. However, we have the constraint that X must be Hermitian, which means we can't choose any arbitrary U. Instead, U must satisfy $U^\dagger = U^{-1}$, which means that U is Unitary. From these two conditions we can state that we have a global $U(N)$ symmetry.

In order to gauge this global symmetry we consider $U = U(\tau)$, so that it is now dependent on Euclidean time. This of course adds an extra complication in our case due to the $\partial_\tau X$ term in the action. We need to ensure that the derivative transforms correctly and that $\partial_\tau(UXU^\dagger) \to U(\partial_\tau X)U^\dagger$. We achieve this through introducing a covariant derivative $D_\tau X$ to solve this issue. We check how the derivative transforms,

$$\begin{aligned}\partial_\tau(UXU^\dagger) &= (\partial_\tau U)XU^\dagger + U(\partial_\tau X)U^\dagger + UX(\partial_\tau U^\dagger)\\ &= U(\partial_\tau X + [U^\dagger(\partial_\tau U), X])U^\dagger.\end{aligned}$$

Where in the above we have used the advantageous fact that $\partial_\tau(UU^\dagger) = \partial_\tau(U^\dagger U) = (\partial_\tau U^\dagger)U + U^\dagger(\partial_\tau U) = 0$. Now we define the covariant derivative as $D_\tau X = \partial_\tau X - i[A, X]$. In this definition we have introduced the gauge field A, which transforms as $A \to UAU^\dagger + iU(\partial_\tau U^\dagger)$. We now have a suitable derivative, $D_\tau(UXU^\dagger) \to U(D_\tau X)U^\dagger$. With this new form for the derivative we have the freedom to directly write the Euclidean action,

$$S_E = \frac{1}{2}\int_0^\beta d\tau \mathrm{Tr}\left[(D_\tau X^i)^2 + (\omega X^i)^2\right]. \tag{2.14}$$

We have not yet discretised the covariant derivative, so to move our analysis to the lattice we must first deal with this continuous term. In this construction we will be following results and methods from [4, 11]. To achieve the discretised case we introduce link variables denoted $g_{n,n'}$ linking lattice site n to n' where $g^{-1}_{n,n'} = g^\dagger_{n,n'} = g_{n',n}$. These variables are formally defined as the path ordered product,

$$g_{n,n+1} = \mathcal{P}\mathrm{exp}\left[i\int_{na}^{na+a} d\tau A(\tau)\right]. \tag{2.15}$$

The visual interpretation of these link variables can be seen in fig.2.2. One can consider thinking of them as a necessary transformation to get from one lattice point to the next, rather than just translation as we had before. This is why in order to describe the derivative as a discrete difference between X_n and X_{n+1} we must first make them comparable using the link variables,

$$D_\tau X \to \frac{g_{n,n+1}X_{n+1}g_{n+1,n} - X_n}{a}. \tag{2.16}$$

We can then use the cyclicity of the trace along with our periodic boundary conditions $X_{\Lambda+1} = X_1$ to rewrite the discrete action in a convenient form,

$$S_{\Lambda,g} = \sum_{n=1}^{\Lambda}\frac{1}{a}\mathrm{Tr}\left\{-X_n\, g_{n,n+1}\, X_{n+1}\, g_{n+1,n} + \left(1 + \frac{\beta^2\omega^2}{2\Lambda^2}\right)X_n^2\right\}. \tag{2.17}$$

It is not ideal that we have multiple $g_{n,n'}$, in fact it would be much nicer to 'squish' the action of all the link variables into one group holonomy, $g = g_{1,2}...g_{\Lambda-1,\Lambda}$. This would simplify our later integrals greatly by allowing us to parametrise the gauge group manifold directly through its invariant Haar measure. We can achieve this by using our previously discussed freedom given by the trace. We choose a specific rotation on X_n,

$$\begin{aligned}&X'_1 = X_1,\ X'_2 = g_{1,2}X_2g_{2,1},\ X'_3 = (g_{1,2}g_{2,3})X_2(g_{1,2}g_{2,3})^\dagger\ldots\\ &X'_\Lambda = (g_{1,2}...g_{\Lambda-1,\Lambda})X_\Lambda(g_{1,2}...g_{\Lambda-1,\Lambda})^\dagger.\end{aligned}$$

Then notice that for $n \neq \Lambda$,

$$\begin{aligned}\mathrm{Tr}\ (X'_nX'_{n+1}) &= \mathrm{Tr}\ \{(g_{1,2}...g_{n-1,n})X_n(g_{1,2}...g_{n-1,n})^\dagger(g_{1,2}...g_{n,n+1})X_{n+1}(g_{1,2}...g_{n,n+1})^\dagger\}\\ &= \mathrm{Tr}(X_ng_{n,n+1}X_{n+1}g_{n+1,n}).\end{aligned}$$

The final result above produces exactly the first term within the action in eq.2.17. This allows us to rewrite the action by taking the $n = \Lambda$ term outside of the sum,

$$S_{\Lambda,g} = \frac{1}{a}\left[-\sum_{n=1}^{\Lambda-1}\mathrm{Tr}\,X'_nX'_{n+1} - \mathrm{Tr}\,X'_\Lambda\, g\, X'_1\, g^\dagger + \left(1 + \frac{\beta^2\omega^2}{2\Lambda^2}\right)\sum_{n=1}^{\Lambda}\mathrm{Tr}\,X'^2_n\right]. \tag{2.18}$$

Finally, we let $g = VDV^\dagger$ where $D = \mathrm{diag}\{e^{i\theta_1}, ..., e^{i\theta_N}\}$, and swap $g \to D$ in eq.2.18. Now we have our final construction for the action, and we can describe the gauged symmetry fully with the eigenvalues of D.

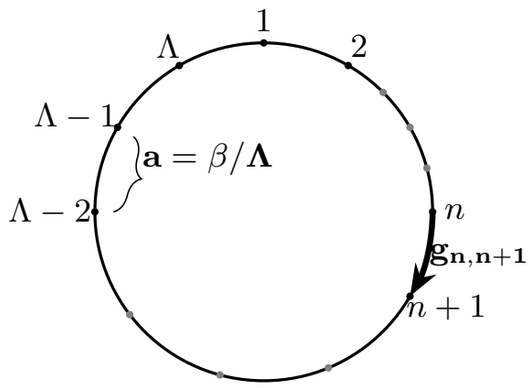


Figure 2.2: Discretisation of the Euclidean time circle into Λ lattice sites. The link variables are shown as a connection from one lattice site to the next.

Now, we return to our discussion of the partition function. To retrace our steps, we first transformed the continuous picture into one on a Euclidean time lattice. We then gauged the $U(N)$ symmetry and introduced the link variables $g_{n,n+1} \in U(N)$, each integrated with the normalised Haar measure $d\mu(g_{n,n+1})$. Then we end up with a new form for the partition function [11],

$$Z_{N,\Lambda} = \int_{U(N)^\Lambda} \int_{\mathbb{R}^{N^2\Lambda}} \prod_{k=1}^{\Lambda} d\mu(g_{k,k+1}) \, \frac{d^{N^2}X_k}{(2\pi\alpha)^{N^2}} \, e^{-S_{\Lambda,g}}. \tag{2.19}$$

The Haar measure is invariant under left and right multiplication, $d\mu(hg) = d\mu(gh) = d\mu(g)$ for any fixed $h \in U(N)$. We have that $g_{\Lambda,1} = (g_{1,2} \cdots g_{\Lambda-1,\Lambda})^{-1} g$, the change of variables $g_{\Lambda,1} \to g$ is a multiplication by a fixed group element, so it has a Jacobian of 1,

$$\prod_{n=1}^{\Lambda} d\mu(g_{n,n+1}) = \prod_{n=1}^{\Lambda-1} d\mu(g_{n,n+1}) \; d\mu(g).$$

After rotating to the X'_n variables the integrand now only has a dependence on g, and we are left a single integral $\int_{U(N)} d\mu(g)$. The integrand is invariant under conjugation so we are allowed to write $g \to V g V^\dagger$, with $D = \mathrm{diag}(e^{i\theta_1}, \ldots, e^{i\theta_N})$ and integrate out V. The integral over the Haar measure is then turned into an integral over the eigenvalue angles [4],

$$d\mu(g) = \frac{1}{N!} \prod_{a=1}^{N} \frac{d\theta_a}{2\pi} \prod_{a<b} \left| e^{i\theta_a} - e^{i\theta_b} \right|^2 \; \propto \; \prod_{a=1}^{N} d\theta_a \; e^{-S_{\mathrm{FP}}[\theta]}, \tag{2.20}$$

$$S_{\mathrm{FP}}[\theta] = -\sum_{l \neq m} \ln \left| \sin \frac{\theta_l - \theta_m}{2} \right|. \tag{2.21}$$

Here the Vandermonde factor $\prod_{a<b} |e^{i\theta_a} - e^{i\theta_b}|^2$ is the Jacobian of the change of variables from matrix elements to eigenvalues and $1/N!$ accounts for permutations of the eigenvalues. The Faddeev–Popov action S_{FP}, gives us an effective repulsion between the eigenvalues. We build a Hamiltonian then by defining conjugate momenta for the X_n and θ_l variables respectively as P_n, Π_l. After these steps the bosonic Hamiltonian becomes,

$$H_{\mathrm{QHO}} = \frac{1}{2} \sum_{n=1}^{\Lambda} \mathrm{tr}\,(P_n P_n) + \frac{1}{2} \sum_{l=1}^{N} \Pi_l^2 + S_{\Lambda,g}[X, D(\theta)] + S_{\mathrm{FP}}[\theta].$$

We will later use this Hamiltonian in our HMC simulations. It is important of course that numerical results are checked where possible against analytical ones. A quick description of how this is achieved is provided by [11] for our particular case which we will summarise. For other cases with different potentials a variety of methods can be found in [21]. One finds that after integrating out the matrices X_n we are left with an integral over the holonomy g of an inverse determinant. In the continuum limit $\Lambda \to \infty$ at fixed $\beta\omega$ this gives,

$$Z_N = \int_{U(N)} \mathcal{D}g \; \frac{e^{-N^2\beta\omega/2}}{\det[\mathbb{1} - e^{-\beta\omega}\, g \otimes g^{-1}]}. \tag{2.22}$$

Notice that the numerator is the zero point energy of the N^2 oscillators. We remove it by normal ordering the Hamiltonian such that,

$$H = \omega \sum_{i=1}^{N^2} \left(a_i^\dagger a_i + \tfrac{1}{2} \right) \to \omega \sum_{i=1}^{N^2} a_i^\dagger a_i,$$

which shifts the energy of every state by $N^2\omega/2$. This construction then gives the Molien–Weyl formula for the generating function of the singlet states, with $x = e^{-\beta\omega}$,

$$Z_N(x) = \int_{U(N)} \mathcal{D}g \; \frac{1}{\det[\mathbb{1} - x\, g \otimes g^{-1}]}. \tag{2.23}$$

Now we can continue as before and diagonalise the holonomy matrix g. In doing this we reduce eq.2.23 to the Cartan torus with eigenvalues $z_i = e^{i\theta_i}$. The Weyl integration formula turns this into N contour integrals around the unit circle,

$$Z_N(x) = \frac{1}{N!}\oint \prod_{i=1}^{N} \frac{dz_i}{2\pi i\, z_i}\, \Delta(z)\Delta(z^{-1}) \prod_{i,j=1}^{N} \frac{1}{1 - x\, z_i z_j^{-1}}, \qquad \Delta(z) = \prod_{i<j}(z_i - z_j), \tag{2.24}$$

which can be evaluated in sequence by residues to give,

$$Z_N(x) = \prod_{n=1}^{N} \frac{1}{1-x^n} = \sum_{n=0}^{\infty} p_N(n)\, x^n, \tag{2.25}$$

where $p_N(n)$ is the number of partitions of n into at most N parts.

Although we have introduced matrix models in a general way initially, defining a system of d $N \times N$ matrices, we have worked with a system where $d = 1$ in this section. When $d \neq 1$ an extra sum over the matrix index $i = 1, \ldots, d$ is added to the action, and each matrix X_n^i has its own set of lattice variables. All d matrices share the same link variables $g_{n,n+1}$ and the same holonomy g, so the gauge integral is still a single integral over g. For the matrix harmonic oscillator, the Gaussian integral over each X^i gives an identical determinant, so if all the masses are equal then the only change is that the determinant is raised to the power d. The more general case is,

$$Z_N(\vec{x}) = \int_{U(N)} \mathcal{D}g \prod_{a=1}^{d} \frac{1}{\det[\mathbb{1} - x_a\, g \otimes g^{-1}]} = \frac{1}{N!}\oint \prod_{i=1}^{N} \frac{dz_i}{2\pi i\, z_i}\, \Delta(z)\Delta(z^{-1}) \prod_{a=1}^{d} \prod_{i,j=1}^{N} \frac{1}{1 - x_a\, z_i z_j^{-1}}, \tag{2.26}$$

where $\vec{x} = (x_1, x_2, \ldots, x_d)$. The extra matrices also only change the HMC simulation through the additional sum in S_b. We will later discuss the BFSS model where a commutator term will couple the different matrices, so this factorisation does not hold and in that case the partition function can't be written in this closed form.

2.3 The $U(N) \times SO(d)$ Gauged Matrix Model

The d matrices of size $N \times N$ transform as a vector under the $SO(d)$ rotations of transverse space. This gives us a global $SO(d)$ symmetry, and it can be checked directly that our action remains invariant under the transformation,

$$X^a \to R_{ab} U X^b U^\dagger.$$

We proceed by gauging this symmetry similarly to how we did with the $U(N)$ case. We will be following [12] and will use results from the $U(N)$ case in section 2.2 to gauge the full $U(N) \times SO(d)$ symmetry. We begin by first introducing the covariant derivative,

$$D_\tau = \partial_\tau + \gamma - i[A, \cdot],$$

where $A(\tau)$ is an an $N \times N$ hermitian matrix and γ is the $SO(d)$ connection. To discretise this expression we continue by once again defining a set of link variables between each lattice site,

$$R_{n,n+1} = \mathcal{P}\exp\left[\int_{na}^{na+a} d\tau \gamma(\tau)\right]. \tag{2.27}$$

The resulting discretised covariant derivative comes out as,

$$D_\tau X \to \frac{R_{n,n+1} g_{n,n+1} X_{n+1} g_{n+1,n} - X_n}{a}. \tag{2.28}$$

We incorporate this covariant derivative directly into the action. Now we have a fully discretised action on a periodic Euclidean time lattice,

$$S_{\Lambda,R,g} = \sum_{n=1}^{\Lambda} \frac{1}{a} \operatorname{tr}\left\{ -X_n\, R_{n,n+1}\, g_{n,n+1}\, X_{n+1}\, g_{n+1,n} + \left(1 + \frac{\beta^2\omega^2}{2\Lambda^2}\right) X_n^2 \right\}. \tag{2.29}$$

We need to look at the partition function once again and retrace our steps. This is necessary to find the additional Faddeev-Popov action terms and to find an analytical way to check results. The introduction of the $SO(d) \times U(N)$ gauge provides us with the following expression,

$$Z_\Lambda(N,d) = \int_{SO(d)^\Lambda} \int_{U(N)^\Lambda} \int_{\mathbb{R}^{dN^2\Lambda}} \prod_{k=1}^{\Lambda} \mu_d(R_{k,k+1})\, \mu_N(g_{k,k+1})\, \frac{d^{dN^2} X_k}{(2\pi a)^{N^2}}\, e^{-S_{\Lambda,R,g}}. \tag{2.30}$$

We can utilise the group invariance and normalization of the Haar measures to reduce the path integral down to a single integration over R and g. We have previously discussed this in section 2.2 for g, and in the case of R we can again use left invariance. The resulting partition function is then,

$$Z_\Lambda(N,d) = \int \mu_d(R)\, \mu_N(g)\, \frac{z_-^{\frac{dN^2\Lambda}{2}}}{\mathbf{det}\left[\mathbb{1} - z_-^\Lambda\, R \otimes g \otimes g^{-1}\right]}. \tag{2.31}$$

Where $z_\pm = 1 + \frac{\alpha^2}{2} \pm \sqrt{\alpha^2\left(1+\frac{\alpha^2}{4}\right)}$, $\alpha = \frac{\omega\beta}{\Lambda}$, and **det** denotes the d-fold matrix determinant. In the continuum limit $\Lambda \to \infty$ at fixed $\beta\omega$ this gives,

$$Z_\infty(N,d) = \int \mu_d(R)\,\mu_N(g)\,\frac{e^{-\frac{dN^2\beta\omega}{2}}}{\mathbf{det}[\mathbf{1} - e^{-\beta\omega}\,R\otimes g\otimes g^{-1}]}. \tag{2.32}$$

Once again the numerator is the zero point energy of the system of QHOs. We let $x = e^{-\beta\omega}$ and we use normal ordering to remove it,

$$\mathcal{Z}(N,d;x) = \int \mu_d(R)\,\mu_N(g)\,\frac{1}{\mathbf{det}[\mathbf{1} - x\,R\otimes g\otimes g^{-1}]}. \tag{2.33}$$

At this point we can diagonalise the holonomy g as we did previously and rewrite the $\mu_N(g)$ part of our partition function as an integral over its eigenvalues,

$$\mathcal{Z}(N,d;x) = \int \mu_d(R)\,\frac{1}{N!}\oint\prod_{i=1}^{N}\frac{dz_i}{2\pi i z_i}\,\Delta(z)\Delta(z^{-1})\prod_{i=1}^{N}\prod_{j=1}^{N}\frac{1}{\mathbf{det}\left[\mathbf{1} - x\,R\,z_i z_j^{-1}\right]}. \tag{2.34}$$

Now we must deal with the $\mu_d(R)$ term. We can block diagonalise R, and using this form we can make it such that all the gauge data is reliant on some variables ϕ_r where $r = 1,\ldots,m$. The block diagonalised form of R provides two answers, one for even d and the other for the odd case,

$$R_{SO(2m)} = \begin{bmatrix} R_2(\phi_1) & & \\ & \ddots & \\ & & R_2(\phi_m) \end{bmatrix}, \qquad R_{SO(2m+1)} = \begin{bmatrix} R_2(\phi_1) & & & \\ & \ddots & & \\ & & R_2(\phi_m) & \\ & & & 1 \end{bmatrix}, \tag{2.35}$$

where in the above,

$$R_2(\phi) = \begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix}. \tag{2.36}$$

We can write R in the block diagonal form and integrate over the angles ϕ_r. In order to do this we must pull the Haar measure back to the maximal torus after which we find [12],

$$\mu_{2m} = \prod_{r=1}^{m}\frac{d\phi_r}{2\pi}\,\frac{\Delta_m^2}{2^{m-1}m!}, \qquad \mu_{2m+1} = \prod_{r=1}^{m}\frac{d\phi_r}{2\pi}\,(2-2\cos\phi_r)\,\frac{\Delta_m^2}{2^m m!}, \tag{2.37}$$

with $\Delta_m = \prod_{r<s}(2\cos\phi_r - 2\cos\phi_s)$. Now we can write the diagonalisation of R in terms of pairs of eigenvalues $\zeta_r = e^{i\phi_r}$. These eigenvalues become the contour measures used in the Molien-Weyl integral, in which the determinant over R factorises into the $\left(1 - x\zeta_r^{\pm 1} z_i z_j^{-1}\right)$ terms of the integrand. For the Monte Carlo simulation we instead exponentiate the Jacobian, exactly as we did for g. Up to constants, $\mu_d \propto \prod_r d\phi_r\, e^{-S_{\rm FP}^{SO(d)}[\phi]}$ with

$$S_{\rm FP}^{SO(2m)}[\phi] = -2\sum_{r<s}\ln\left|2\cos\phi_r - 2\cos\phi_s\right|, \tag{2.38}$$

$$S_{\rm FP}^{SO(2m+1)}[\phi] = S_{\rm FP}^{SO(2m)}[\phi] - \sum_{r=1}^{m}\ln\left(2-2\cos\phi_r\right). \tag{2.39}$$

Defining momenta Φ_r, the Hamiltonian gains a kinetic term for these angles and the new Jacobian term,

$$H = \frac{1}{2}\sum_{n,i}\mathrm{tr}\left(P_n^i P_n^i\right) + \frac{1}{2}\sum_{l=1}^{N}\Pi_l^2 + \frac{1}{2}\sum_{r=1}^{m}\Phi_r^2 + S_{\Lambda,R,g}[X, D(\theta), R(\theta)] + S_{\rm FP}^{U(N)}[\theta] + S_{\rm FP}^{SO(d)}[\theta]. \tag{2.40}$$

Back to discussing the analytical case, we actually are able to write a solvable form for the partition function of the $U(N)\times SO(d)$ gauged system of QHOs. These two solutions are separated by an even and odd case, for $SO(2m)$ we have,

$$\mathcal{Z}(N,2m;x) = \int \mu_{2m}(\zeta)\,\mu_N(z)\prod_{r=1}^{m}\prod_{i,j=1}^{N}\frac{1}{\left(1 - x\zeta_r z_i z_j^{-1}\right)\left(1 - x\zeta_r^{-1} z_i z_j^{-1}\right)}, \tag{2.41}$$

while for $SO(2m+1)$ we get

$$\mathcal{Z}(N,2m+1;x) = \int \mu_{2m+1}(\zeta)\,\mu_N(z)\prod_{i,j=1}^{N}\frac{1}{\left(1 - x z_i z_j^{-1}\right)}\prod_{r=1}^{m}\frac{1}{\left(1 - x\zeta_r z_i z_j^{-1}\right)\left(1 - x\zeta_r^{-1} z_i z_j^{-1}\right)}. \tag{2.42}$$

These expressions can be evaluated exactly for reasonable N and m. They can also be solved through expansions in x, for which the coefficients in the series expansion can be found via contour integration. We now have a descriptions for the matrix QHO system with both $U(N)$ and $U(N)\times SO(d)$ gauged. This concludes our discussion of the matrix QHO, we will now move to discussing the situation where the Gaussian potential in the action is replaced by a squared commutator term. This produces the bosonic BFSS action.

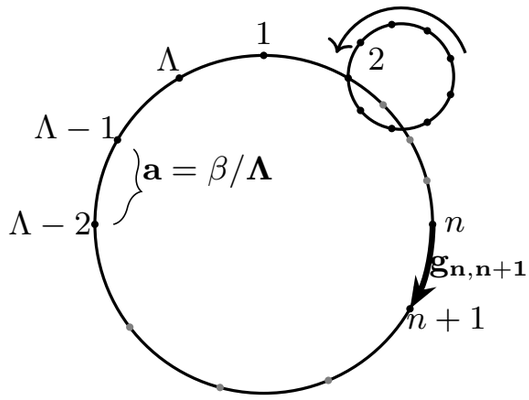


Figure 2.3: Discretisation of the Euclidean time circle into Λ lattice sites. The link variables are shown as a connection from one lattice site to the next, and the freedom for $SO(d)$ rotation is represented like a rotation of the nine matrices at a given lattice site as a vector.

3 The Bosonic BFSS Model

The Bosonic part of the BFSS action is given as,

$$S_E = \frac{1}{g^2}\int_0^\beta d\tau\, \mathrm{Tr}\Big\{\tfrac{1}{2}(D_\tau X^i)^2 - \tfrac{1}{4}[X^i, X^j]^2\Big\} \tag{3.1}$$

The only difference between our action now and the Gaussian case we have outlined in detail in section 2 is the potential term. Rather than $(X_n^i)^2$ we now have a commutator squared potential. In fact, when $d \to \infty$ the interacting matrix model admits a Gaussian reduction [21]. Although this does complicate things by mixing matrices, for our purposes it is not much different from what we have already looked at. For reasons previously discussed in section 2.2 and 2.3, there is a $U(N) \times SO(9)$ global symmetry for this model. The discretised actions, both gauged for $U(N)$ and for $U(N) \times SO(d)$, will come out the same bar the potential. The contributions to the Hamiltonian via Faddeev-Popov actions will also be the same, so it is simple to upgrade our HMC simulation to the bosonic BFSS case. For example, the discretised $U(N)$ gauged bosonic BFSS action will come out as,

$$S_{\Lambda,g} = \frac{1}{g^2}\sum_{n=1}^{\Lambda} a\,\mathrm{Tr}\Big\{\frac{1}{2}\Big(\frac{g_{n,n+1}X^i_{n+1}g_{n+1,n} - X^i_n}{a}\Big)^2 - \frac{1}{4}[X^i_n, X^j_n]^2\Big\} \tag{3.2}$$

$$= \frac{1}{2g^2 a}\sum_i\Big[\sum_{n=1}^{\Lambda-1}\mathrm{Tr}(X^i_{n+1} - X^i_n)^2 + \mathrm{Tr}\big(DX^i_1 D^\dagger - X^i_\Lambda\big)^2\Big] - \frac{a}{4g^2}\sum_{n=1}^{\Lambda}\mathrm{Tr}[X^i_n, X^j_n]^2. \tag{3.3}$$

For the $U(N) \times SO(d)$ case we have a similar result to eq.3.2 where $X^{n+1} \to R_{n,n+1}X_{n+1}$. In the case of the BFSS model the commutator term makes finding a general closed form solution for the partition function intractable. It is shown in [21] that in the large d limit this model becomes a $U(N)$ gauged Gaussian action via Gaussian reduction, for which the partition function can be evaluated in closed form using a Molien–Weyl integral. It was also shown that in the low temperature limit the bosonic singlet spectrum is organized into BFSS_2 like towers associated with the quadratic Gram operators $\mathrm{Tr}(X^a X^b)$. We won't discuss this in much detail in this report, as in general for our simulation we will be considering systems with $N = 10,\ d = 9$.

3.1 The Orientifold

In order to project the BFSS action onto the orientifold, we must impose new constraints on the X_n^i matrices. The orientifold involves a reflection of the transverse spatial coordinates, $\vec{x} \to -\vec{x}$, which induces a projection on the matrix degrees of freedom. The matrices are then restricted to be either symmetric or antisymmetric depending on their transformation under the orientifold, [9] satisfying,

$$X = \pm(X^T). \tag{3.4}$$

We choose the antisymmetric case. Our matrices are now antisymmetric as well as hermitian, so they are purely imaginary. This means we have,

$$(X_n^i)^\dagger = X_n^i, \quad (X_n^i)^T = -X_n^i, \qquad X_n^i \to R^T X_n^i R, \quad R \in SO(N)$$

This implies that we must have $RR^T = \mathbb{I},\ \det(R) = 1$. Nothing changes to our original $SO(d)$ symmetry in this case. Thus the $U(N)$ symmetry is replaced by $SO(N)$, while the original $SO(d)$ rotational symmetry remains. This means we can examine the $SO(N)$ gauged case and then later examine the $SO(N) \times SO(d)$ gauged case. When we gauge the full $SO(N) \times SO(d)$ symmetry we are able to pick up two Faddeev-Popov action terms which are added to our HMC Hamiltonian,

$$H = \tfrac{1}{2}\sum_{n,i}\mathrm{tr}(P_n^i P_n^i) + \tfrac{1}{2}\sum_r \Pi_r^2 + \tfrac{1}{2}\sum_r \Psi_r^2 + S_\Lambda[X, g(\theta), R(\phi)] + S_{FP}^{SO(N)}[\theta] + S_{FP}^{SO(d)}[\phi]. \tag{3.5}$$

Where these actions are given by,

$$S_{FP}^{SO(2m)}[\phi] = -2\sum_{r<s}\ln|2\cos\phi_r - 2\cos\phi_s|, \qquad S_{FP}^{SO(2m+1)}[\phi] = S_{FP}^{SO(2m)}[\phi] - \sum_r \ln(2 - 2\cos\phi_r). \tag{3.6}$$

The steps by which we accomplished the above have been discussed in sections 2.2 and 2.3.

4 Observables

In this section we will define some thermodynamic quantities with on which we will focus our analysis. A natural choice to begin with is the energy E, where we write the standard definition from statistical mechanics,

$$E/N^2 \equiv -\frac{1}{N^2}\frac{\partial \ln Z}{\partial \beta}. \tag{4.1}$$

A similar quantity to the energy of the system is the extent of space, which is often used in analysis of matrix models as its form is not computationally heavy. The expectation value of the extent of space is given as,

$$\langle R^2 \rangle \equiv \left\langle \frac{1}{N^2\beta}\int_0^\beta d\tau \ \mathrm{Tr}(X^i X^i) \right\rangle. \tag{4.2}$$

The expectation value of the extent of space is expected to behave the same as the energy for the Gaussian matrix harmonic oscillator by the virial theorem (in the large Λ limit). In the case of the bosonic BFSS system the energy given as,

$$\frac{E}{N^2} = -\frac{3}{4g^2N^2\Lambda}\sum_{n=1}^{\Lambda}\sum_{i,j}\left\langle \mathrm{Tr}[X_n^i, X_n^j]^2 \right\rangle \tag{4.3}$$

The last observable that we look at is the Polyakov loop, and it can be found for any gauged group. We define it as,

$$P \equiv \frac{1}{N}\ \mathrm{Tr}(g),\ \ g \equiv \mathcal{P}\mathrm{exp}\left(i\int_0^\beta dt A(t)\right). \tag{4.4}$$

Where in the above case we've used g to denote the holonomy for the $U(N)$ gauge. Due to the trace it is clear that the above can be calculated then as a sum of phases since,

$$\mathrm{Tr}(g) = \ \mathrm{Tr}(VDV^{-1}) = \ \mathrm{Tr}(D) = \sum_{i=1}^{N} e^{i\theta_i}$$

. The $SO(d)$ and $SO(N)$ cases comes out differently due to the diagonalisation of the holonomy. In that case we have,

$$P_{SO(d)} \equiv \frac{1}{d}\mathrm{Tr}(R) \tag{4.5}$$

For $SO(d)$, the eigenvalues of the holonomy occur in complex-conjugate pairs, $e^{i\phi_1}, e^{-i\phi_1}, \ldots, e^{i\phi_m}, e^{-i\phi_m}$, with an additional eigenvalue 1 for odd $d = 2m+1$. Thus,

$$\mathrm{Tr}(g_{SO(2m)}) = 2\sum_{a=1}^{m}\cos\phi_a, \tag{4.6}$$

$$\mathrm{Tr}(g_{SO(2m+1)}) = 1 + 2\sum_{a=1}^{m}\cos\phi_a, \tag{4.7}$$

and hence the Polyakov loop can be evaluated directly from the holonomy eigenphases. To determine the possible point at which a phase transition may occur in the Polyakov loop we define the Polyakov loop susceptibility as,

$$\chi_P = \left(\langle |P|^2\rangle - \langle |P|\rangle^2\right). \tag{4.8}$$

This is the quantity we will be using throughout our description of our results to determine possible critical temperatures for any system we look at. In general for the $SO(N)$ or $U(N)$ gauge we define the critical temperature as T_c^{colour}, whereas for the $SO(d)$ gauge we write $T_c^{SO(d)}$. It is important to note that although we refer to $T_c^{SO(d)}$ as a "critical temperature" for convenience, it is not necessarily indicative of a phase transition in the same sense as T_c^{colour}.

5 Results

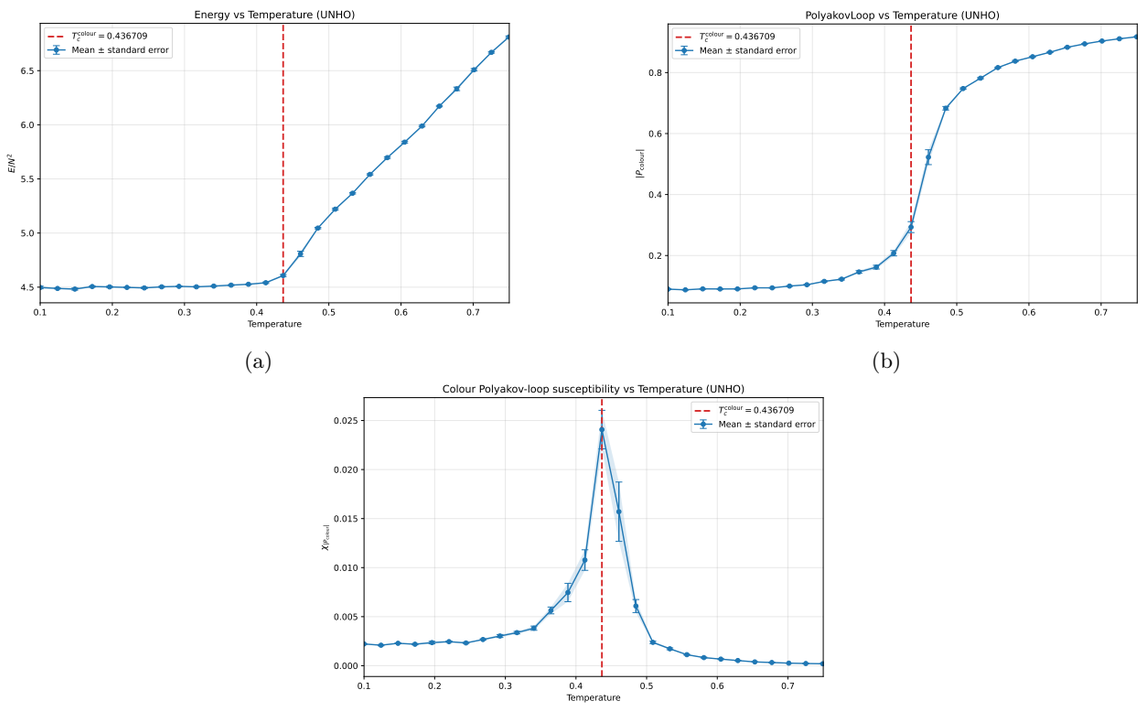
In this section we will run through and discuss our results from our HMC simulations. In general we have set our system up to have $d = 9$ and $N = 10$. This is the highest N we could reasonably achieve but it may still be considered quite small so we expect some finite N error in our results. We also use a variable lattice length Λ which is often determined by thermalisation time. In general we also set $\omega = 1$, and for the BFSS model we have $\lambda = g^2N = 1$.

Due to time constraints we weren't able to produce exact estimates for the error in our results for critical temperature, though we do show a standard error for the data points generated in the presented figures. We try to compare our results to known quantities where possible.

5.1 The Quantum Harmonic Oscillator

We begin with the discussion of the $U(N)$ gauged QHO system. In this case thermalisation came nicely, and the Polyakov loop susceptibility in fig.5.1c shows a somewhat clear peak. From this result we were able to get a relatively confident estimate for the critical temperature $T_c^{\text{colour}} \approx 0.44$. The supposed phase transition can be seen taking place at this temperature in figures 5.1a and 5.1b, where we can see a change in the regimes for both relationships. If one were to zoom in on the energy versus temperature plot, it is clear to observe the same shape as the Polyakov loop relationship around the critical temperature. Energy however is linear with respect to temperature after this point, unlike the Polyakov loop.

We can verify if these results are reasonable by looking at known results for this system as seen in [5]. We find that we should expect a critical temperature in the large N limit as $T_c^{\text{colour}} = \frac{\omega}{\ln(d)} = \frac{1}{ln(9)} \approx 0.46$. Our critical temperature at $T_c^{\text{colour}} \approx 0.44$ gives us an error of around 4.34%. The relationships exhibited in figures 5.1a and 5.1b also match those expected in [5]. We find that indeed P has taken a value between 0 and 1/2 as expected at $T = T_c^{\text{colour}}$ with $P \approx 0.3$. This suggests that at the critical temperature we should find $E/N^2 = d/2 + P^2 \approx 4.59$. This is indeed very close to our resulting value at $E/N^2 \approx 4.6$.

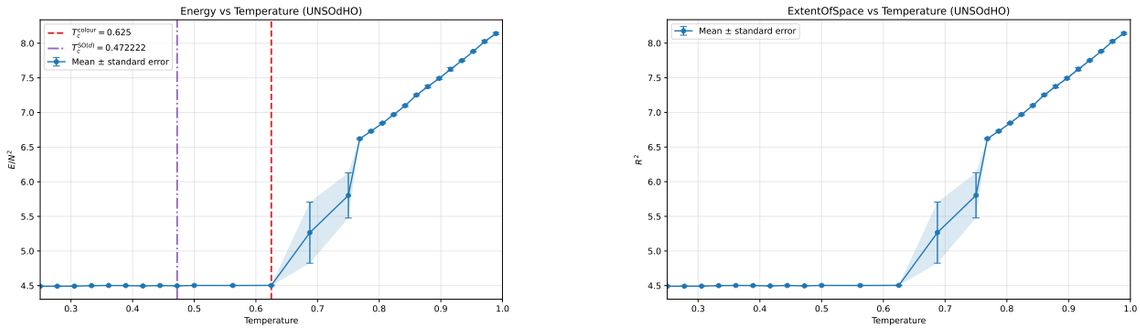


(a) (b)

(c)

Figure 5.1: All figures demonstrate results for observables on the $U(N)$ gauged QHO. In figure (a) we see energy plotted as a function of temperature with the approximate critical temperature T_c^{colour} indicated on top to make the phase transition clear. Figure (b) exhibits a similar phase transition in the Polyakov loop where we can see the same relationship as (a) except there is no linear dependence after the transition. In figure (c) we show how we got our estimate for the critical temperature using the peak of the Polyakov susceptibility.

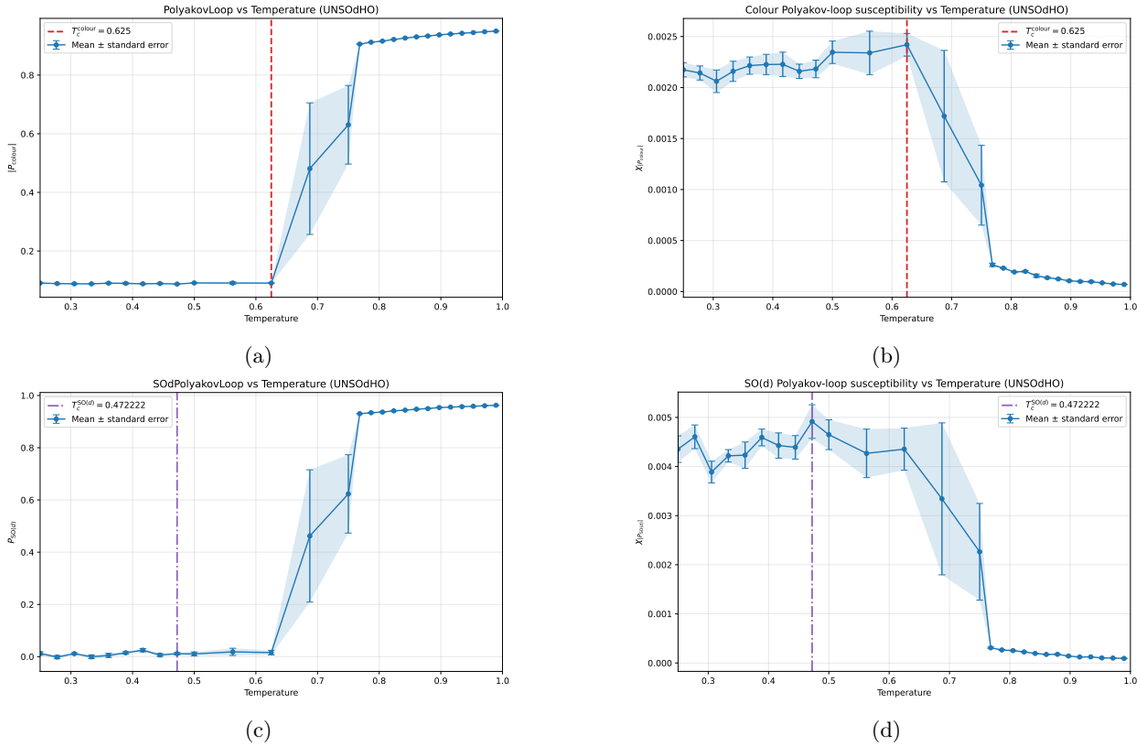
Moving to the $U(N) \times SO(d)$ system, thermalisation in this case took much longer. Due to the long thermalisation time our results have higher error than the previously examined case. This high error however seems to be localised around the actual phase transition point as seen in figure 5.2, the low and high temperature limits are better behaved in general. In the Gaussian model the extent of space R^2 and energy E/N^2 should be in agreement with each other (in the high Λ limit) which is also exactly what we see in this figure.



(a) (b)

Figure 5.2: Both figures demonstrate the behaviour of observables in relation to the $U(N) \times SO(d)$ QHO. In figure (a) we have shown the relationship between energy and temperature with two results for critical temperatures indicated over this plot. In figure (b) we have shown the extent of space which is expected to exhibit the same relationship as figure (a) in the Gaussian system.

We examined the Polyakov loop for both the $U(N)$ and $SO(d)$ groups separately. The critical temperature for the $U(N)$ gauge was found to be roughly $T_c^{\text{colour}} \approx 0.63$, and for the $SO(d)$ gauge we found $T_c^{SO(d)} \approx 0.47$. The error on these approximations is of course very high, with no truly exact peak exhibited in figures 5.3b and 5.3d. For this reason it is more accurate to describe them in terms of a window of temperatures based on the observed jump in the Polyakov loop, $T_c^{\text{colour}}, T_c^{SO(d)} \in [0.5, 0.8]$. Despite these large errors at the phase transition once again the high and low temperature limits are relatively well behaved and produce expected results. Both Polyakov loops tend to 1 as expected.



(a) (b)

(c) (d)

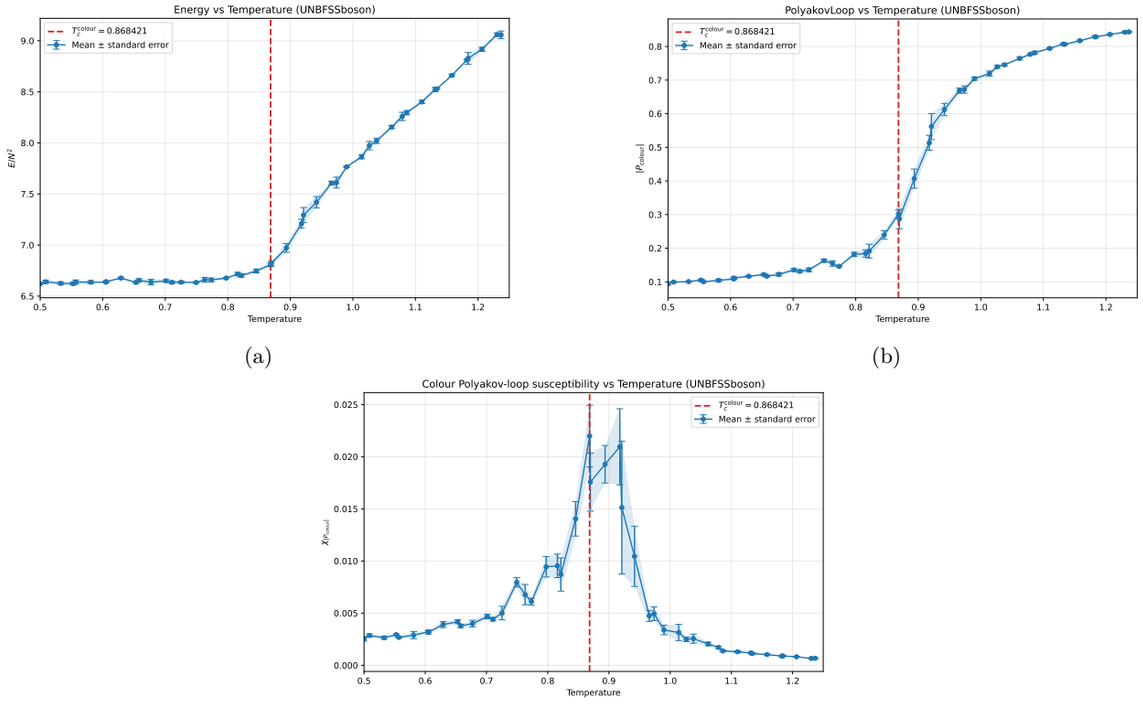
Figure 5.3: All figures are in relation to the $U(N) \times SO(d)$ QHO system. Figures (a) and (b) show results for the Polyakov loop for the $U(N)$ gauge, with (b) exhibiting how the critical temperature is gained from the Polyakov susceptibility. Figures (c) and (d) follow similarly where (d) shows how the estimate for the critical temperature was obtained.

5.2 The Bosonic BFSS

The $U(N)$ gauged bosonic BFSS model has been widely studied across literature. In our case we see much of what has been seen before. The thermalisation time is a bit longer than the Gaussian case likely due to the coupled matrices introducing some extra complexity which did affect the error of our critical temperature estimate. From the peak in susceptibility as seen in figure 5.4c we have roughly $T_c^{\text{colour}} \approx 0.87$. It has been previously shown in [8] that a critical temperature of $T_c^{\text{colour}} \approx 0.9$ is expected from this system. Our results seem to be in good agreement with this approximation. We have also reproduced roughly the same relationships in figures 5.4b and 5.4a as seen in [8]. For our system we have a lower $N = 10$ rather than their $N = 16, 32$ and we do not see quite as sharp of an incline in the Polyakov loop.

The high-temperature behaviour is qualitatively consistent with the leading-order expansion found in [7], in which the non-zero Matsubara modes decouple and the zero modes are governed by the bosonic IKKT-type matrix model. This predicts $\langle R^2 \rangle \approx \chi_1 T^{1/2}$, $\langle P \rangle \approx 1 - \frac{1}{2d}\chi_1 T^{-3/2}$ and $E/N^2 \approx \frac{3}{4}\chi_2 T$, so that $1 - \langle P \rangle$

falls off as $T^{-3/2}$ while the energy grows linearly in T.

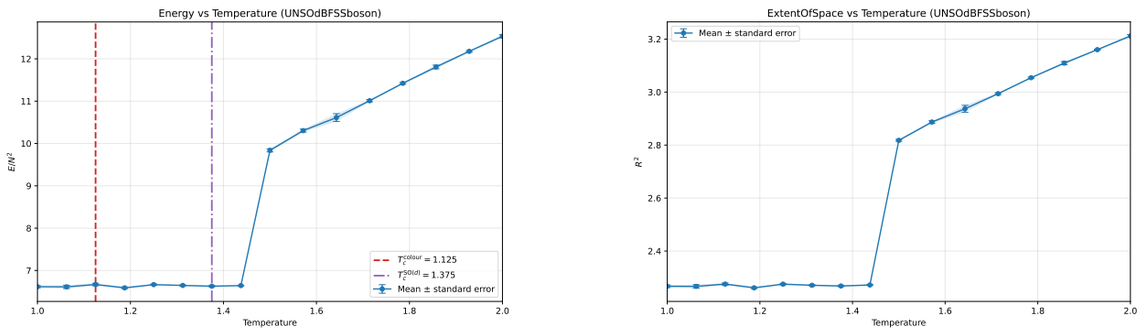


(a) (b)

(c)

Figure 5.4: All figures concern the bosonic BFSS model with $U(N)$ gauge. In figure (a) we see the energy versus temperature relationship exhibit the expected shape, with a phase transition occurring at roughly $T_c^{\text{colour}} \approx 0.868421$. Figures (b) and (c) concern the Polyakov loop where (c) exhibits a peak in the susceptibility at the supposed critical temperature.

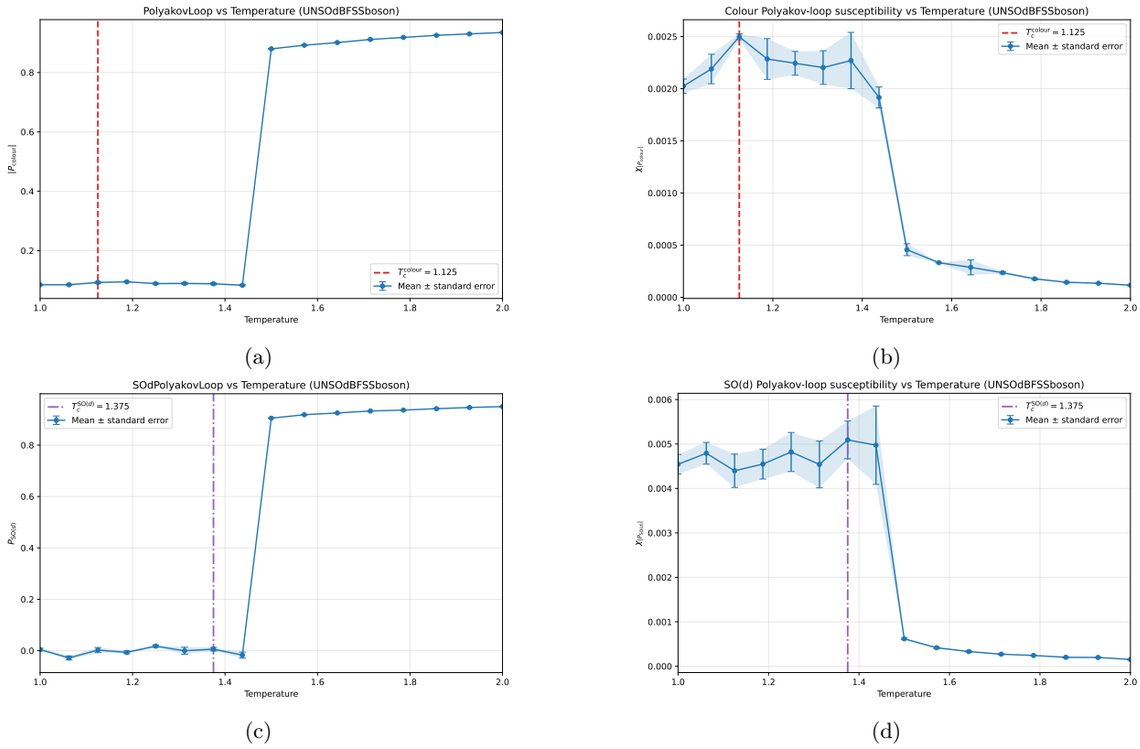
Now we turn our discussion to the $U(N) \times SO(d)$ gauged bosonic BFSS model. In this case the thermalisation times similarly to the QHO do get worse once again, and error becomes higher. As seen in figure 5.5 the low and high temperature regimes are well behaved while the area in which the phase transition occurs is not. The extent of space and energy relationships with temperature seem to agree, although they begin at very different values in the low temperature limit. Since we have the commutator term in the action the extent of space and the energy are not expected to exactly agree.



(a) (b)

Figure 5.5: Both figures are in relation to the bosonic BFSS system with $U(N) \times SO(d)$ gauged. In figure (a) we can see the energy versus temperature relationship, and in figure (b) we see the extent of space versus temperature relationship. Both tend to behave very similarly, especially in the low and high temperature regimes.

We couldn't find any results for a similarly sized system for critical temperature values to compare to. It is still useful however to consider the Hagedorn temperature $T_H = \frac{1}{ln(d)} \approx 0.4551$ [12]. In the Gaussian case this temperature is expected to be associated with the Hagedorn transition. We can use T_H as a reference temperature when examining the behaviour of our observables. We measured that $T_H << T_c^{\text{colour}} < T_c^{SO(d)}$, with $T_c^{\text{colour}} \approx 1.13$, $T_c^{SO(d)} \approx 1.38$ As can be seen in figures 5.6b and 5.6d the estimates for these critical temperatures have very high errors. Once again it is more accurate to describe these with a window based on the observed jump in the Polyakov loop and we see that T_c^{colour}, $T_c^{SO(d)} \in [1.1, 1.5]$. We also note that this window is still higher than the results from the $U(N)$ gauge on its own, so the critical temperature has likely risen.

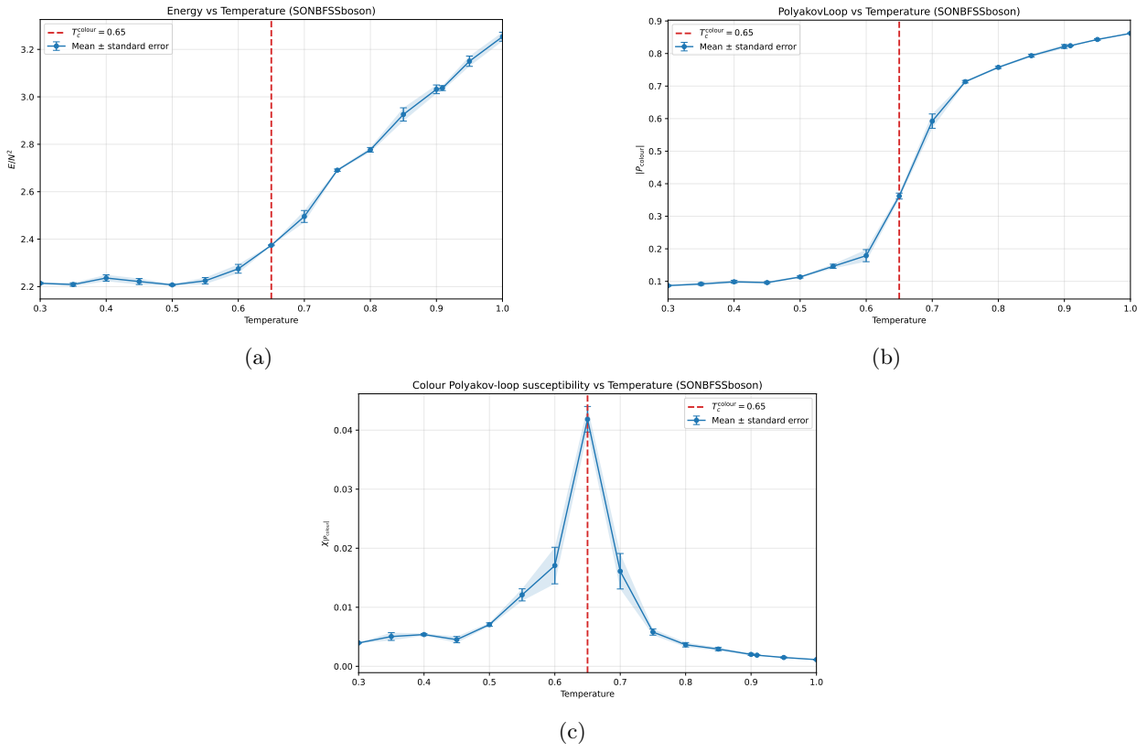


(a) (b) (c) (d)

Figure 5.6: All figures are in relation to the $U(N) \times SO(d)$ gauged bosonic BFSS model. The Polyakov loops exhibited the expected relationship with temperature, however their susceptibility is not very clear in figures (b) and (d). The $SO(d)$ Polyakov loop in figure (a) starts at a much lower value than that of the $U(N)$ case in figure (c) yet both return to the same point in the high temperature regime later.

5.3 The Bosonic BFSS on the Orientifold

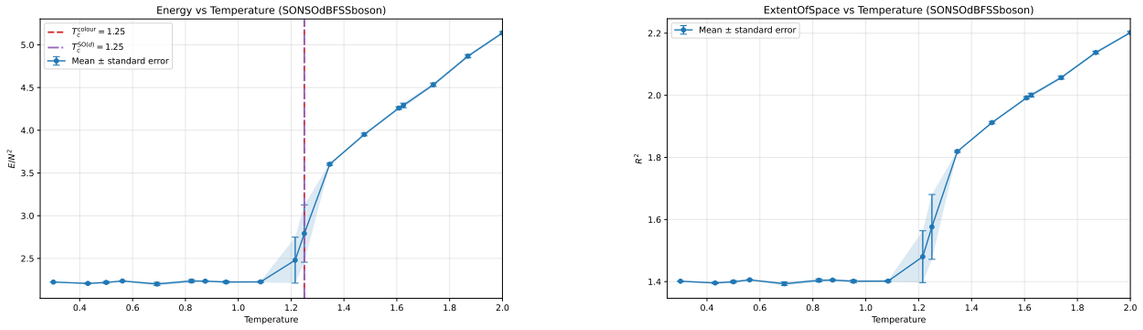
The bosonic BFSS projected onto the Orientifold forces the X_n^i to be antisymmetric and gives us a $SO(N)$ symmetry. Gauging this symmetry produced comparable results to the $U(N)$ gauged bosonic BFSS system, as can be seen in figure 5.7, however the critical temperature we measured at $T_c^{\text{colour}} \approx 0.65$ is quite a bit lower. This system seemed to thermalise nicely so the Polyakov loop susceptibility successfully exhibits a visible peak in figure 5.7c. The low-temperature value of E/N^2 is lower for $SO(N)$ at ≈ 2.2 than for $U(N)$ at ≈ 6.6. This is consistent with the smaller number of degrees of freedom in the $SO(N)$ case.



(a) (b) (c)

Figure 5.7: All figures here are for the $SO(N)$ gauged bosonic BFSS on the orientifold. In general we see the same energy temperature relationship in figure (a) that we've seen for previously discussed cases. Figure (b) shows the Polyakov loop also exhibits the previously seen relationship however the incline at which the phase transition occurs is not as extreme. In figure (c) we see that the estimate for the critical temperature does have a visible peak in the Polyakov loop susceptibility.

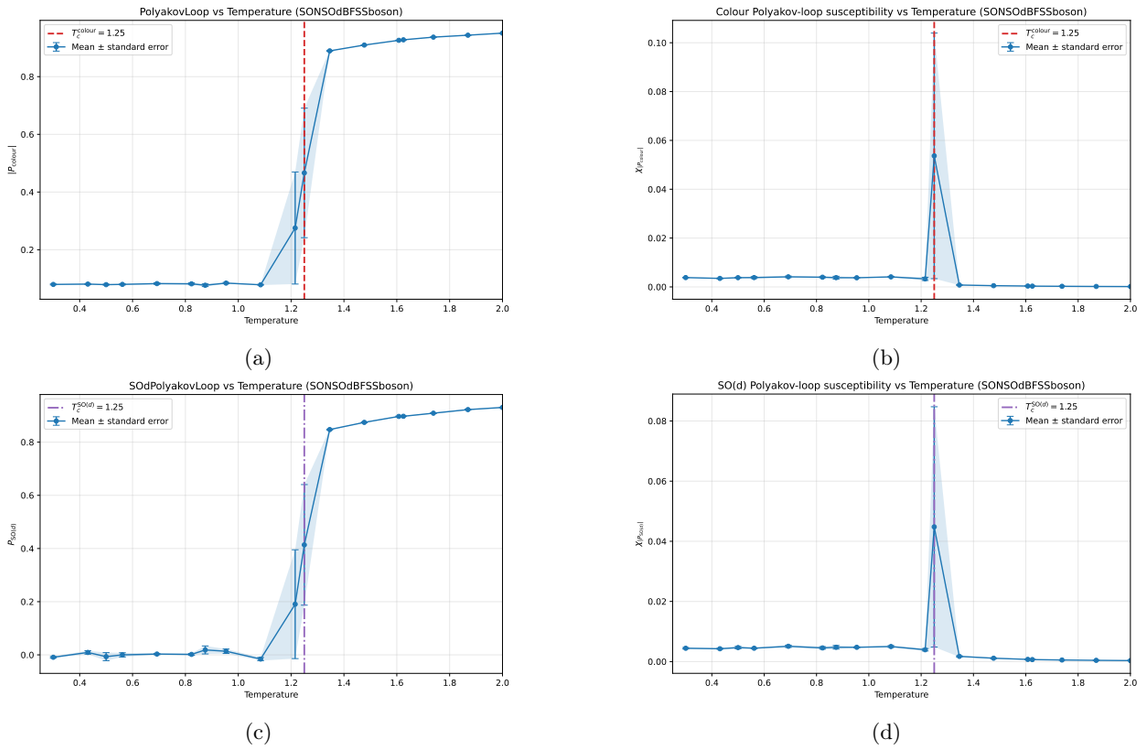
The $SO(N) \times SO(d)$ gauged system produced the expected relationships that we've previously seen for energy and extent of space measured against temperature in figure 5.8. Similarly to the $U(N) \times SO(d)$ case for the bosonic BFSS system we see that the critical temperatures have become much higher than with just the $SO(N)$ or $U(N)$ gauge alone.



(a) (b)

Figure 5.8: These observables are for the $SO(N) \times SO(d)$ bosonic BFSS on the orientifold. Figures (a) and (b) exhibit the same relationship roughly as a function of temperature, where (a) concerns energy and (b) extent of space.

Since in previous cases the thermalisation time became much longer once an $SO(d)$ gauge was introduced we expected the same result here. The error on the estimates for critical temperature did indeed get worse which can be seen in figures 5.9b and 5.9d, however a clear peak is still visible. Our measured critical temperatures coincide within error at $T_c^{\text{colour}} \approx T_c^{SO(d)} \approx 1.25 = T_{\text{orientifold, c}}$.



(a) (b)

(c) (d)

Figure 5.9: These observables are for the $SO(N) \times SO(d)$ bosonic BFSS on the orientifold. We observe a visible peak in the Polyakov loop susceptibility in figures (c) and (d). These estimates are shown to follow through in the behaviour of the Polyakov loop in figures (a) and (b).

6 Conclusion

We have used Hamiltonian Monte Carlo methods to study thermodynamic quantities of gauged matrix models on a Euclidean lattice. We began with the matrix harmonic oscillator and extended our discussion to the bosonic BFSS model including its orientifold projection. In each case the gauge degrees of freedom were reduced to the holonomy eigenphases. This gave us additional Faddeev-Popov terms in the action. Possible transition temperatures were estimated from peaks in the Polyakov loop susceptibilities of the colour and $SO(d)$ holonomies.

For the $U(N)$ gauged Gaussian Hamiltonian we found $T_c^{\text{colour}} \approx 0.44$ which is in good agreement with the expected large-N Hagedorn temperature $\frac{1}{\ln(9)} \approx 0.46$. The $U(N)$ gauged bosonic BFSS system was similarly well behaved and we found $T_c^{\text{colour}} \approx 0.87$ which is also in good agreement with previous results of $T_c^{\text{colour}} \approx 0.9$ in [8]. The introduction of the $SO(d)$ gauge in the Gaussian and bosonic BFSS models

produced much larger statistical error around the phase transition, although the low and high temperature limits remained well behaved. For the $SO(N)$ gauged bosonic BFSS projected onto the orientifold we obtained $T_c^{\text{colour}} \approx 0.65$. Once the $SO(d)$ gauge was introduced the errors did once again become larger but we did find that the critical temperature from the Polyakov loop of both gauges came out to be consistent within error, $T_c^{\text{colour}} \approx T_c^{SO(d)} \approx 1.25$. One reason why thermalisation time may have taken so much longer upon the introduction of the $SO(d)$ gauge is that the Faddeev-Popov terms produce large forces when eigenphases approach one another. This might have called for a smaller step size. Due to time constraints we were unable to test this hypothesis.

The biggest hurdle we faced throughout this project was small matrix size and struggling with thermalisation times. We were unable to run the simulation in some cases for long enough to produce accurate values. In future work we would include more error analysis, particularly for finite Λ and N, such that these results can be presented with accurate uncertainties. We would also aim to look at the areas around the phase transition temperature with a much finer temperature grain.

The direction for future work would certainly be to make this model fully supersymmetric. This would be accomplished through the inclusion of fermionic terms, which would need to be handled with Rational HMC methods. Along with this we would be interested in the thermodynamic quantities produced by the BMN mass deformation case. Finally one of the quantities that would have also been interesting to consider is heat capacity as investigated in [12]. The analysis performed in this report is in relation to the canonical ensemble which strictly concerns non negative heat capacity. If we were to do a detailed investigation between different heat capacity branches in the microcanonical ensemble, we could obtain further insight into the phase structure of the model.

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Appendix A: Hybrid Monte Carlo Methods

Throughout this report we use Hamiltonian Monte Carlo (HMC) methods to determine thermodynamic quantities. In this section we will give a brief explanation of the particular algorithm we used. Much of the discussion here can be found in [10], which has a more detailed description of the method and its use cases.

In Quantum field theory (QFT) we use euclidean path integrals to define a partition function,

$$Z = \int \mathcal{D}x e^{-S_E[x]}. \tag{A.1}$$

This in turn gives us a probability distribution $P(x) = \frac{1}{Z} e^{-S_E[x]}$ which is useful as it allows us to turn the calculation of expectation values into a problem of sampling from a probability distribution. There are a variety of Monte Carlo (MC) methods to choose from that make this sampling easily done on a computer. Monte Carlo methods are algorithms that use random sampling to estimate quantities. Markov Chain Monte Carlo (MCMC) methods are a type of Monte Carlo method in which the samples are generated by a Markov chain, such that each new state depends only on the current state and the chain converges to the desired probability distribution. One of the most well known examples of an MCMC algorithm is that of Metropolis Hastings. In this method firstly a new sample is proposed based on the previous sample, after this the proposed one is either added to the set of samples or rejected based on the value of the probability distribution at that point. Algorithm 1 demonstrates the use of Metropolis Hastings on the probability distribution derived from eq.(A.1).

Algorithm 1 Metropolis–Hastings update

Store $x_i = x_{\text{old}}$
$x = x_i + \delta x$
$\Delta S = S(x) - S(x_i)$
if $\exp(-\Delta S) > \text{Rand}[0, 1]$ **then**
    $x_{i+1} = x$
    successes = successes + 1
else
    $x_{i+1} = x_i$
end if

This is, although useful, not very efficient. We want to build a system that will maximise the number of accepted samples without clumping them together. We want to cover as much ground as possible but we don't want to waste computation. One of the ways we can modify Algorithm 1 to better suit our needs is to make it aware of the physical evolution of the system. To do this we introduce the Leap Frog Algorithm 2, which is a method by which we can integrate the Equations of Motion (EoM).

The Hamiltonian formulation of mechanics gives us a set of EoM based on a given phase space. The phase space is defined with position coordinates q_i and conjugate momentum coordinates p_j. The EoM come out as,

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \tag{A.2}$$

$$\dot{p}_i = -\frac{\partial H}{\partial q_i}. \tag{A.3}$$

The Leap Frog Algorithm uses these equations to evolve q_i and p_i in time. We begin at some initial position $q_0 = q(0)$ and momentum $p_0 = p(0)$. Then we introduce some time step ϵ. We can then find $q(2\epsilon)$, $p(2\epsilon)$ by performing the steps in Algorithm 2, where we have defined $F_i(x) = -\dot{p}_i = \partial H/\partial q_i$. Algorithm 2 combines two leapfrog steps of size ϵ, merging the two adjacent half step momentum updates into one full step. Iterating this method m times up to $t_f = 2m\epsilon$ will allow us to find $q(t_f)$ and $p(t_f)$.

Algorithm 2 Leapfrog integrator

Input: p_0, x_0, ϵ
$p = p_0 - \frac{\epsilon}{2} F(x_0)$
$x = x_0 + \epsilon p$
$p = p - \epsilon F(x)$
$x = x + \epsilon p$
$p = p - \frac{\epsilon}{2} F(x)$
Return: x, p

So, in our HMC algorithm we want to use the method of Metropolis Hastings which uses $P(q) = \frac{1}{Z}e^{-S_E}$ as a probability distribution and also use the Leap Frog Algorithm to produce better proposals. We can start merging these two ideas by first introducing a fictitious Hamiltonian $H = S[q] + \frac{1}{2}p^2$. This makes no change to our probability distribution other than in the normalisation factor because the momenta can be integrated out directly $P(q) \propto \int dp, e^{-H(q,p)} = e^{-S_E[q]} \int dp, e^{-p^2/2} \propto e^{-S_E[q]}$. This means that we can use this newly defined H to generate equations of motion and to define a probability distribution $P_H(q,p) = \frac{1}{Z_H}e^{-H(q,p)}$. The HMC algorithm we used in this report is given in Algorithm 3.

Algorithm 3 Hybrid Monte Carlo update

$p_i = \mathcal{N}(0,1)$
Store $x_i = x_{\text{old}}$, $p_i = p_{\text{old}}$
$(x,p) = \text{Leapfrog}(x_i, p_i)$
for $j = 1, \ldots, m-1$ **do**
    $(x,p) = \text{Leapfrog}(x,p)$
end for
$\Delta H = H(x,p) - H(x_i, p_i)$
if $\exp(-\Delta H) > \text{Rand}[0,1]$ **then**
    $x_{i+1} = x$
    $\text{successes} = \text{successes} + 1$
else
    $x_{i+1} = x_i$
end if

The next question is how to determine a suitable step size ϵ and number of Leapfrog steps m. We would like the acceptance rate to be at roughly 60 % if possible such that we cover enough ground with reasonable samples. There's a multitude of adaptive techniques that will optimise ϵ for this purpose. One of these is the dual averaging method which can adapt the step size during an initial tuning phase towards the desired acceptance rate. In our report this was the solution we chose to this issue, in particular adapting Algorithm 5 given in [6].